

Real-Time On-Device Voice Spoof & Synthetic Call Detection

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AI tools can now clone a person's voice using just a short 3-second audio recording. Scammers are using these cloned voices to impersonate police officers, bank officials, or family members to steal money--a major issue right now with "Digital Arrest" scams in India.

- Sending call audio over the internet is too slow to warn someone in real time.
- It requires an internet connection and leaks private conversations.
- It drains laptop battery and can make video calls laggy.

To give users a physical warning they cannot miss, VoiceArmor-Edge also connects via USB to an Arduino UNO Q board. If a fake voice is detected, the Arduino immediately turns on a bright red warning light and buzzer.

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The Problem

- Voice Cloning Fraud: Fraudsters call victims pretending to be law enforcement or family members in distress, demanding urgent bank transfers.
- No Protection on Meeting Apps: Popular apps like Microsoft Teams, Zoom, and WhatsApp Desktop have no built-in way to verify whether a voice is human or AI-generated.

- Fast Response: Checking speech on the NPU takes less than 15ms, warning the user while the caller is still speaking.
- Complete Privacy: Audio never leaves the laptop. Everything works 100% offline, even in Airplane Mode, keeping conversations completely private.
- No Video Call Lag: The NPU handles the AI calculations, leaving the CPU and GPU free to handle the video call smoothly.
- Low Battery Drain: The NPU uses very little power, allowing the protection to run quietly in the background all day.

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System Flow Diagram

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? - Applies Hanning window (smooths edge noise) ?
? - Fast Fourier Transform (STFT) ?
? - Produces 64-band Mel-Spectrogram (frequency image) ?
??
? Spectrogram Tensor [1, 1, 64, 96]
?
??
? 2. Qualcomm AI Hub Model ?
? - Compact audio classification CNN backbone ?
? - Converted to INT8 (8-bit integers) for fast math ?
? - Compiled with Qualcomm AI Hub tools ?
? - Targeted for Snapdragon X chips ?
??
? Optimized Model
?
??
? 3. Snapdragon HP PC Runtime ?
? - Runs on Windows using ONNX Runtime with QNN plugin ?
? - Connects directly to Hexagon NPU via QnnHtp.dll ?
? - Inference time: under 15ms per chunk ?
? - Produces a live spoof probability score (0 to 1) ?
??
? Score > 0.85 (Fake Voice)
?
??
? 4. Instant User Alerts ?
? - On Screen: Windows notification alert ?
? - On Hardware: Sends signal over USB to Arduino ?
? - Turns on bright red warning LED ?
? - Activates warning buzzer ?
??

Key Technical Components

A. Audio Feature Extraction (Digital Signal Processing)

- When AI tools generate fake speech, they leave subtle irregularities in higher frequencies that natural human vocal cords do not produce.
- The application captures audio from your computer's audio output in real time.
 - It slices the sound into 250ms chunks and generates a Log-Mel Spectrogram (a visual representation of sound frequencies over time).
 - This step runs quickly on the CPU and hands the data straight to the NPU.

B. Qualcomm AI Hub Model

- The classifier is adapted from standard audio models in the Qualcomm AI Hub catalog.
- INT8 Quantization: Model weights are converted from 32-bit floating points to 8-bit integers. This speeds up processing on the Hexagon NPU and uses very little memory with virtually no drop in detection accuracy.
- The model is compiled using Qualcomm's official `qai-hub` toolchain to run as an ONNX model with Qualcomm QNN acceleration.

C. Running on Windows on Snapdragon

- The app runs as a native Windows service using `onnxruntime` with the `onnxruntime-qnn` execution provider plugin.
- The model links directly with Qualcomm's `QnnHtp.dll` driver, routing tensor computations straight to the Hexagon NPU cores.

D. Hardware Alert with Arduino UNO Q

In line with the Snapdragon AI Lab setup:

- The Snapdragon HP PC does the heavy AI computation.
- The Arduino UNO Q connects via USB to provide a physical, out-of-band indicator.
- If a fake voice is detected, the PC signals the Arduino to turn on an external red warning LED and buzzer.
- Even if the user has their screen covered, minimized, or full-screened, the physical light immediately alerts them to hang up.

4. Why This Approach Stands Out

1. Practical & Focused: Instead of trying to do too many things at once, it solves one critical, high-impact problem very well: live synthetic

voice detection.

2. True Edge Computing: Proves why an on-device NPU is necessary. Cloud APIs simply cannot match the sub-15ms speed and privacy required here.
3. Physical Hardware Safety: Combining the HP PC with the Arduino UNO Q bridges software AI with real-world physical hardware alerts.

5. Deployment on Snapdragon HP PCs

- Target Device: HP OmniBook laptops powered by Snapdragon X Elite / X Plus.
- Operating System: Windows 11 on ARM64.
- Lightweight: Uses less than 120 MB of RAM and under 3% CPU.
- Works Offline: Operates completely in Airplane Mode without any internet connection.
- Easy to Use: Runs quietly in the Windows system tray and monitors call audio automatically.

6. Project Presentation & Code

- Slide Deck: A clean 10-slide presentation explaining the problem, architecture, NPU integration, and future roadmap.
- Source Code: Full project code on GitHub, including the audio processing pipeline (`voicearmor_dsp.py`), NPU inference script (`qnn_inference.py`), and Arduino sketch (`voicearmor_alert.ino`).

7. Conclusion

VoiceArmor-Edge is a focused, realistic, and practical project that shows the real value of the Snapdragon Hexagon NPU. By combining digital signal processing, Qualcomm AI Hub model optimization, and a physical hardware alert, it protects everyday users from growing AI voice scams safely and privately on their own laptops.